

Data Visualization Project

Milestone 2

1 Project Goal

The goal of this project is to design an interactive tool that helps users better understand nutritional information and make informed food choices.

The platform is built around two complementary entry points. On one hand, users can provide personal inputs such as dietary preferences (e.g., low sugar, vegetarian) or a target calorie intake, allowing the system to highlight food options adapted to their needs. On the other hand, users can directly explore the available food items without specifying constraints, and access detailed nutritional information for each of them.

From a user perspective, the website acts as a flexible decision-making tool. Users can examine individual foods, understand their nutritional composition, and, most importantly, compare different options in a meaningful way. Comparison plays a key role, as it allows users to go beyond isolated values and evaluate trade-offs between foods. For instance, they can quickly understand whether one food is higher in protein but also higher in calories, or identify alternatives that better match their objectives.

These objectives will be achieved through a set of complementary and comparative visualizations, designed to support both detailed analysis and direct comparison between food items. The main visualizations used in this project are presented in the following section.

2 Visualizations

To help users explore and understand the data, we use complementary visualizations that provide both detailed and global insights into nutritional information.

2.1 Radar chart (nutritional composition and comparison)

The radar chart displays the nutritional profile of a selected food item across multiple dimensions - proteins, carbohydrates, fats, fiber, and sugars - encoded as the area of a polygon on a set of radial axes. When a second food is selected, a second polygon is overlaid, enabling direct visual comparison of two items, making it instantly clear which food has higher or lower values for each nutrient.

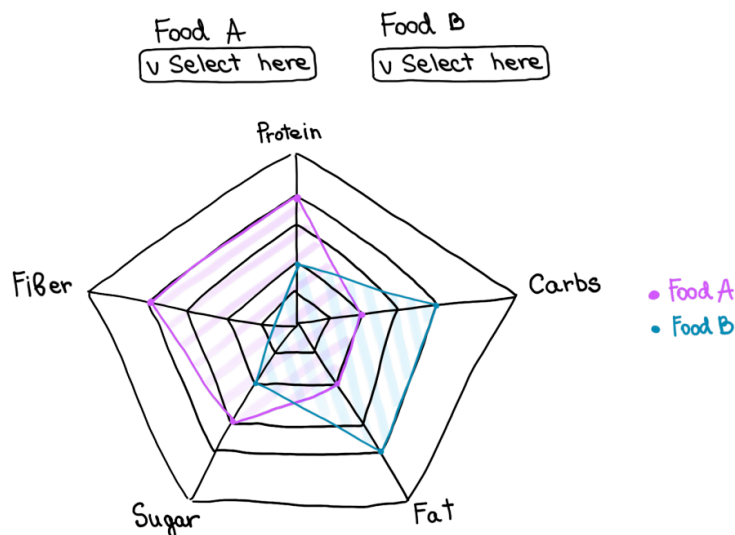


Figure 1: Radar chart sketch: two food profiles overlaid for comparison

2.2 Parallel coordinates chart (global overview of foods)

The parallel coordinates chart provides a global view of the food dataset. Each food item is represented as a polyline connecting its values across several vertical axes, each corresponding to a nutritional attribute (proteins, carbohydrates, fats, sugars, fiber, calories). This encoding also allows users to detect patterns and clusters within the dataset.

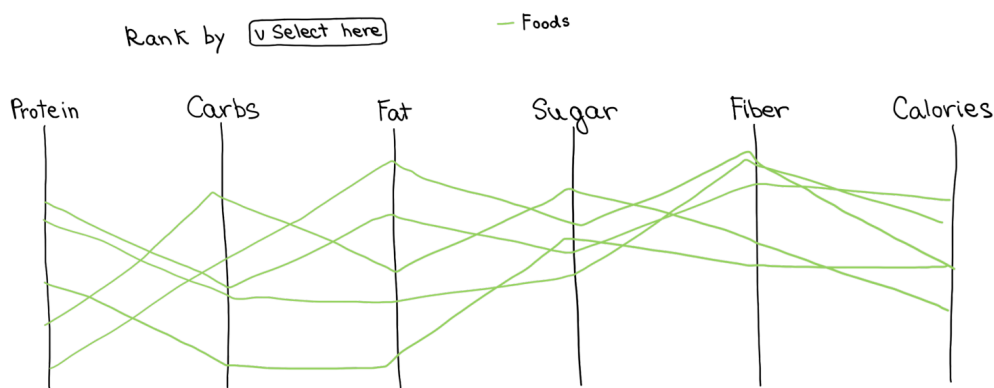


Figure 2: Parallel coordinates chart sketch: each line is a food item

2.3 Pie chart (recommended plate composition)

The pie chart shows the recommended macronutrient distribution for a single meal - proteins, carbohydrates, and fats - as proportional arc segments. The proportions are computed from the user's Total Daily Energy Expenditure (TDEE), personal goal, and dietary preferences, and adapted to a per-meal basis. This visualization serves as a personalised nutritional reference, giving users a clear target to compare against the foods they explore in other views.

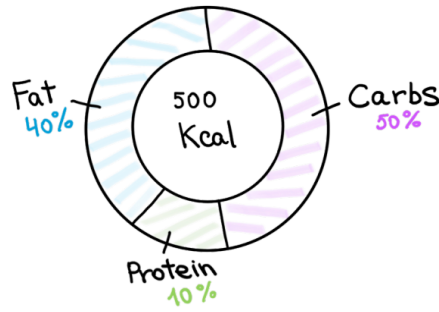


Figure 3: Pie chart sketch: macronutrient distribution personalised to the user's goal for a single meal

2.4 Scatterplot (explore all foods at once)

The scatterplot encodes each food item as a dot positioned according to its caloric (x-axis) and protein (y-axis) content, allowing users to immediately identify nutritionally efficient foods - high in protein relative to their calorie count. Dietary filters (low sugar, vegetarian, gluten-free, etc.) and a calorie slider allow users to restrict the visible items. Clicking a dot selects the corresponding food. The top-left region of the chart is highlighted as the ideal zone, visually guiding users towards foods that best combine nutritional density with caloric efficiency.

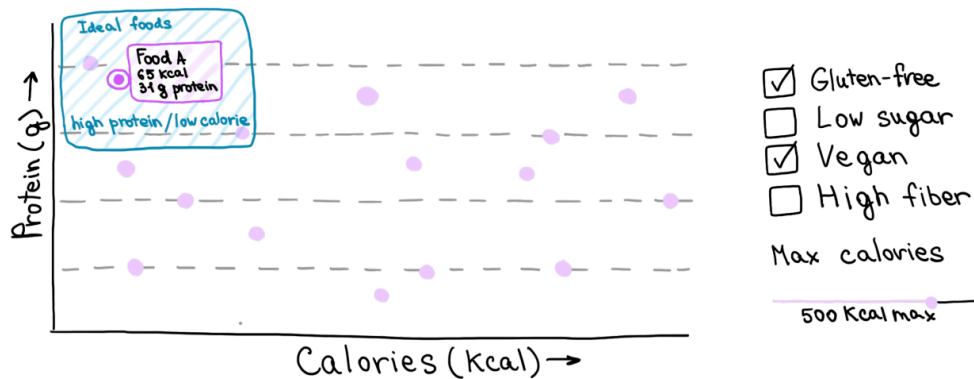


Figure 4: Scatterplot sketch: each dot is a food item; the highlighted zone marks the ideal protein-to-calorie ratio

3 Tools and Course Material

To develop our project, we rely on the web technologies and visualization concepts introduced throughout the course.

On the technical side, we use HTML, CSS, and the DOM to structure the interface, JavaScript (ES6) to handle interactivity, and D3.js to create interactive visualizations. Data preprocessing is carried out using Python and Pandas.

Our approach is guided by several key lectures, including *JavaScript (Lectures 2 and 3)* and *D3.js (Lecture 4)* for implementation, as well as *Data, Interactions*, and *More Interactive D3.js* for handling dynamic user interactions. We also rely on *Marks and Channels* and *Perception and Colors* to design clear and effective visual encodings, and on *Designing Visualization* to ensure readability and usability.

4 Advanced Features

Beyond the basic functionalities, we want to make the system more useful by adding features that make it more personalized and easier to use.

- The system will take into account the user's daily energy needs using the concept of Total Daily Energy Expenditure (TDEE), which corresponds to the number of calories a person burns in a day. By allowing users to compute or input this value, we can use it as a reference to adapt recommendations and visualizations.
- Users will be able to define a daily calorie goal depending on their objective, such as maintaining their weight, losing weight, or gaining muscle. Based on this goal, the system will generate a recommended nutritional balance and display it through visualizations such as a recommended plate and adapted comparisons between food items.
- Dietary restrictions and preferences will be integrated into the system. Users will be able to specify constraints such as allergies or intolerances (e.g., gluten-free, lactose-free) or dietary choices (e.g., vegetarian). The visualizations will update dynamically to only display foods that match these conditions.
- The system will suggest alternative foods that satisfy both the nutritional targets and the selected constraints, allowing users to identify suitable options more easily.
- The system will also provide simple explanations to help users understand their choices, for example by highlighting why a food is more suitable than another depending on protein, fat, or sugar content.

5 Prototype

We developed an initial prototype of our website that already demonstrates how users will interact with the system.

It is available at: <https://com-480-data-visualization.github.io/BossLadies/>

The idea is to guide the user from personal input to data exploration. Users can first enter basic information in the TDEE calculator (age, weight, height, activity level). This is how we will estimate daily calorie needs and use them later for personalization.

In the Explore Food Information section, users can select different foods and compare them. We implemented a radar chart to compare the nutritional composition of selected foods (proteins, carbohydrates, fats, etc.), and a parallel coordinates chart to visualize multiple foods at once and identify patterns across the dataset.

The prototype also includes a first version of the nutrition database. Foods will be later organized into categories such as meats, vegetables, and grains. This allows users to browse and select items more easily, and will be connected to filtering options.

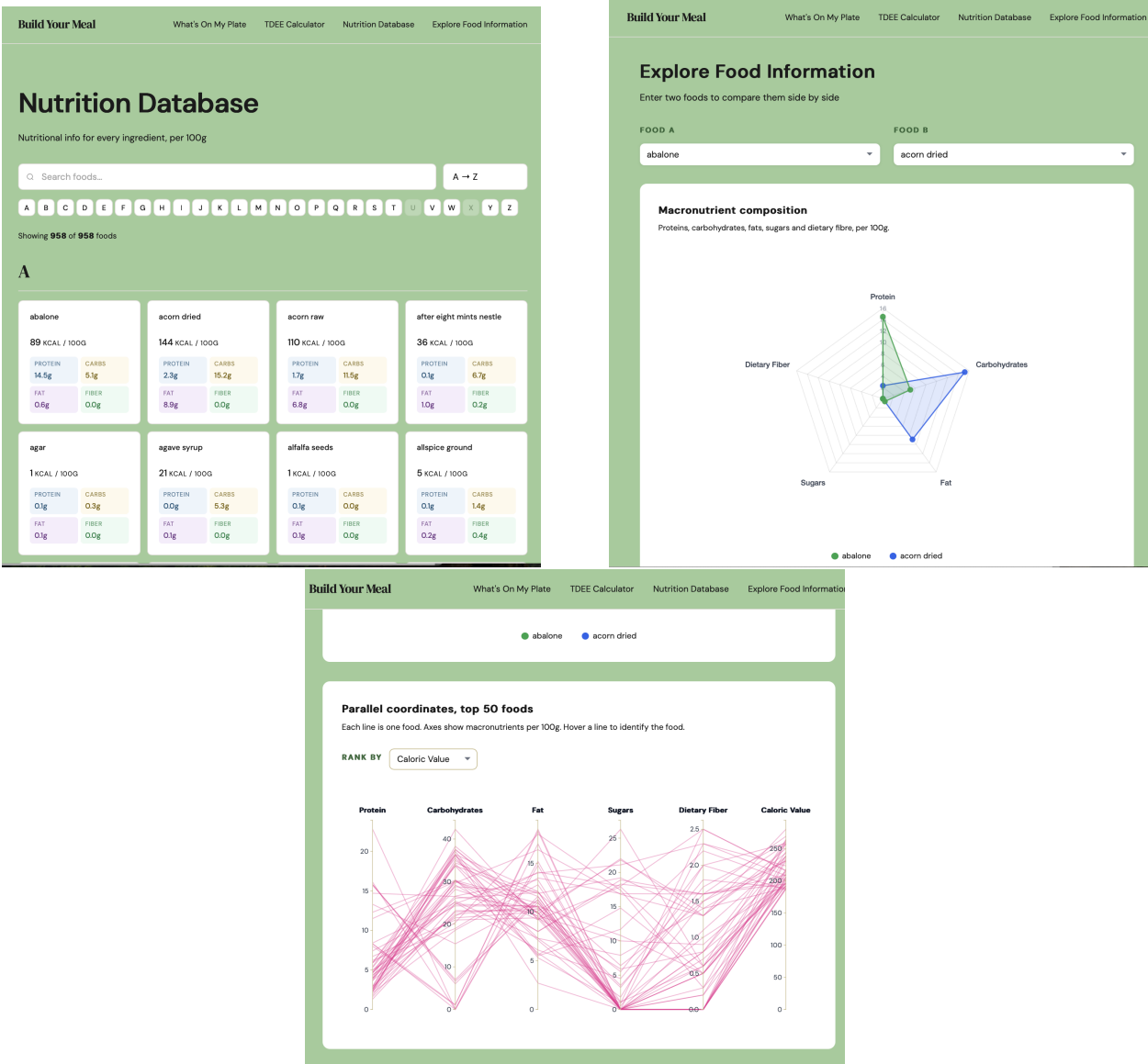


Figure 5: Prototype of the website